MILITARY SPECIFICATION

TRACTOR, WHEELED; 16 HORSEPOWER, HYDROSTATIC DRIVE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a minimum 16 horsepower (hp), gasoline-engine-driven, hydrostatic drive, four-wheel tractor.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-T-3351 - Tractor, Full-Tracked, Low-Speed, Tractor Wheeled, Agricultural, and Tractor Wheeled, Industrial: and Their Attachments, Packaging of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2420

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), the contractor shall furnish a tractor for first article inspection and approval (see 4.2.1 and 6.3).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3 Identical items. All units of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.4 Safety. Parts subject to high operating temperatures, moving or rotating parts, including rear wheels, and parts energized electrically which are of such a nature or so located as to become a hazard to operating and maintenance personnel shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions.

3.5 Speed, dimensions, and weight. Tractor shall have a maximum travel speed of not less than 6.5 miles per hour (mph) forward and 2.5 mph reverse. The tractor shall conform to the dimensions specified in table I. Shipping weight, without accessories, shall be not less than 675 pounds (lb). The maximum drawbar pull without ballast shall be not less than 390 lb.

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TABLE I. Dimensions.

Item	Dimensions (in.) Min	Max.
Overall dimensions:		
Length	61	85
Width	34	44
Height	37	48
Wheelbase	43	56

3.6 Frame. Frame shall be fabricated of structural steel with members joined using one of the arc-welding processes. Frame shall be mounted on two axles equipped with wheels. Wheels shall be mounted on ball or roller type bearings.

3.7 Tires. Manufacturer's standard tires shall be provided.

3.8 Fenders. Rear wheel fenders shall be provided.

* 3.9 Brake systems. Systems to satisfy emergency service and parking brake requirements shall be provided.

3.10 Drawbar hitch. Manufacturer's standard fixed type drawbar hitch shall be provided.

* 3.11 Engine. Tractor shall be powered by a four cycle, air cooled, gasoline engine. The corrected maximum brake hp of the engine shall be determined at the engine manufacturer's rated speed, but shall not exceed 3,600 revolutions per minute (rpm). The corrected maximum brake hp shall be not less than 16, and shall be determined in accordance with SAE J607. Engine shall be equipped with a governor, set to limit engine speed to the maximum operating speed recommended by the engine manufacturer. The tractor's shipping weight to corrected maximum brake hp shall be not greater than 62 p und/horsepower. The tractor shall be capable of performing satisfactorily in any ambient temperature from +110° degree Fahrenheit (°F) -20°F. The engine shall be furnished complete with the following:

(a) A fuel tank with a capacity of not less than 3 gallons, and equipped with a fuel indicator.
(b) Carburetor intake air cleaner.
(c) Muffler with exhaust pipe located to assure expelled gases are directed away from the operator. Vertical exhaust pipes shall be equipped with self-opening and closing rain caps.
(d) A 12-volt electrical cranking system with battery. The battery shall be 12-volt potential, negative ground, and have a capacity of not less than 39 ampere-hours at the 20-hour rate. Battery shall be shipped charged and dry. Electrolyte shall not be furnished.
(e) A starter control and key type ignition switch.
(f) A battery charging generator or alternator.
(g) An ammeter or indicator light.
(h) Manual speed control and choke.
(i) An hour meter.

* 3.12 Power train. Transmission of power from the engine to the rear drive wheels shall be accomplished using a hydrostatic transmission. When operating

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on a smooth, level, hard surface, transmission shall transmit sufficient engine power to the drive wheels to maintain the tractor at a constant ground speed of not less than 6.5 mph forward and 2.5 mph reverse. The transmission shall have a medium output torque range.

3.13 Steering. Tractor shall be equipped for front wheel steering. An automotive type steering mechanism shall be provided. The steering mechanism shall be either hydraulically power-assisted or the steering ratio shall be reduced.

3.14 Lighting. Tractor shall be furnished with two headlights and a floodlight. The floodlight shall be mounted on the tractor in a position which shall allow the light beam to be directed toward the hitch specified in 3.11. Floodlight shall be provided with an amber lens and a separate operating switch.

3.15 Operator's station. Operator's station shall be equipped with a seat and backrest, instrument panel, and operating controls. Seat and backrest shall be padded and covered with weatherproof material. The seat shall be horizontally adjustable. Instrument panel shall contain the instruments except hour meter, controls, and switches specified in 3.11 and 3.14. All instruments except fuel indicator, shall be visible and all hand and foot operated controls, including brake, throttle, transmission, and steering wheel, shall be within easy reach of the operator while sitting in his normal operating position. Instruments, gages, switches, and controls shall be protected against moisture and inclement weather conditions. Each instrument, gage switch, and control shall be provided with permanently affixed identification markings.

3.16 Accessories and allied equipment.

3.16.1 Cab. When specified (see 6.1), manufacturer's standard fully enclosed all metal cab shall be provided.

3.16.2 Reel mower. When specified (see 6.1), manufacturer's standard rear mounting 3-gang reel mower shall be provided.

3.16.3 Snow blower. When specified (see 6.1), manufacturer's standard 40-inch minimum plowing width, front mounted, quick detachable snow blower shall be provided.

3.17 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication.

3.18 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit. Nonferrous screws, rivets, or bolts of not less than 1/8 in in diameter shall be used to affix the plates. Nomenclature shall be "TRACTOR, WHEELED, HYDROSTATIC DRIVE."

* 3.19 Instruction plates. The tractor shall be equipped with manufacturer's standard instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment.

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3.20 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of finish coat. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.21 Workmanship.

3.21.1 Steel fabrication. Steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.21.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.21.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.21.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one tractor when a first article is required (see 3.1 and 6.1, and 6.3). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each tractor shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. When a first article is required, the first article shall be subjected to the tests specified in 4.4.1 and 4.4.2, and when equipped with attachments, to 4.4.3. Failure to pass any phase of the tests shall be cause for rejection.

4.4.1 Road test. Tractor shall be lubricated, fueled, and weighed. All necessary adjustments shall be made, including the seat. Tractor shall be observed to determine conformance to:

a. The weight requirement of 3.5.
b. The seat adjustment requirement of 3.15.
c. The lubrication requirements of 3.17.

The weight to hp ratio shall be determined to verify compliance with 3.11. Tractor shall perform the maneuvers as specified in table II and shall be tested to demonstrate conformance to the requirements of 3.12, 3.13, and 3.14.

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TABLE II. Test requirements.

Type maneuver	Direction	Speed (mph)	Time (min)
Travel	Forward	1 to 1.5	30
	Forward	2.5 to 3.5	30
	Forward	6.5 to max.	30
	Reverse	0.5 to 1.5	15
	Reverse	1.5 to 2.5	15
Figure eights	Forward	1 to 6.5	45

4.4.2 Drawbar test. To determine conformance to 3.5, the tractor shall be tested for maximum drawbar pull without ballast in accordance with SAE J708. The test shall be conducted on a clean, level concrete surface.

4.4.3 Allied equipment test. When a 3-gang reel mower or snow blower is furnished, each piece of equipment shall be mounted on the tractor and operated for at least 30 minutes.

4.4.4 Production sample. Upon acceptance of the first article, it shall remain at the manufacturing facility as a production sample, and be the last tractor shipped on the contract The first artice shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.5 Quality conformance tests. Each production tractor shall be completely assembled, properly adjusted, and serviced for operation. Engine shall be started and subjected to a warmup period as recommended by the manufacturer. Each tractor shall perform one-third of the maneuvers as specified in table II. All controls, including light switches, shall be operated as many times as necessary to determine ease of operation, effectiveness, responsiveness, and that mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or excessive vibration. The hydrostatic transmission, steering system, bearings, shafts, and other moving parts shall be tactually examined for excessive heating or abnormal operation insofar as is practicable. Failure to pass any phase of this test shall be cause for rejection.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the quality conformance inspection of MIL-T-3351.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.1), equipment shall be preserved and packaged in accordance with the mobile requirements (type II) of MIL-T-3351 with the level of preservation and packaging as specified (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. When a first article is required for inspection and approval (see 3.1, 4.2.1, and 6.3).
c. When fully enclosed all metal cab is required (see 3.16.1)
d. When a rear mounted 3-gang reel mower is required (see 3.16.2).
e. When a snow blower is required (see 3.16.3).
f. Color of finish coat required (see 3.20).
g. When the preservation and packaging requirements of MIL-T-3351 are required and the level of preservation and level of packing required (see 5.1).

* 6.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete tractor. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-29188A(YD)	TRACTOR, WHEELED; 16 HORSEPOWER, HYDROSTATIC DRIVE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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